

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

- A) Draw and explain the neat block diagram of fundamental steps in Digital Image Processing. (BT2/CO1) 6
- B) Consider the image segment shown in the figure that follows. (BT3/CO1) 6
Let $V = \{0,1\}$ be the set of intensity values used to define adjacency. Compute the lengths of the shortest 4-, 8-, and m -path between p and q in the following image.
If a particular path does not exist between these two points, explain why.

	3	1	2	1 (q)
	2	2	0	2
	1	2	1	1
(p)	1	0	1	2

Repeat it using $V = \{1,2\}$.

- C) Formulate and explain the distance measures between two pixels are (any three). (BT2/CO1) 6

Q.2 Solve Any Two of the following.

- A) Define gray-level intensity transformation. Elaborate methods of gray level intensity transformation. (Linear, Logarithmic & Power law correction). (BT2/CO2) 6
- B) Compute histogram equalization of the image. (BT4/CO2) 6

4	5	6	5	4
5	6	4	5	5
4	6	4	5	5
4	5	6	4	5
6	4	5	6	5

- C) Compute DCT Transform on (BT3/CO2) 6

$$f(x, y) = \begin{bmatrix} 2 & 4 & 4 & 2 \\ 4 & 6 & 8 & 3 \\ 2 & 8 & 10 & 4 \\ 3 & 8 & 6 & 2 \end{bmatrix}$$

Q. 3 Solve Any Two of the following.

A) Write a note on Boundary Extraction and Hole Filling. (BT1/CO3) **6**

B) Find Global thresholding for given image. (BT3/CO3) **6**

5	3	9
2	1	7
8	4	2

C) Compute opening and closing of A & B. (BT4/CO1) **6**

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \& \quad B = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Q.4 Solve Any Two of the following.

A) State and explain Gradient based edge detection algorithm with example. (BT2/CO4) **6**

B) What is corner detection? Explain The Harris and Stephens algorithm for corner detection. (BT2/CO4) **6**

C) Describe neighborhood with HOG features. (BT3/CO4) **6**

Q. 5 Solve Any Two of the following.

A) Explain pattern and pattern classes. (vector, string and trees) (BT2/CO5) **6**

B) Describe pattern classification by prototype matching. (BT2/CO5) **6**

C) Explicate matching structural prototypes based on matching shape numbers. (BT3/CO5) **6**

***** End *****